**Impact** Since being online, the bAbI tasks have already directly influenced the development of several promising new algorithms, including weakly supervised end-to-end Memory Networks (MemN2N) of Sukhbaatar et al. (2015), Dynamic Memory Networks of Kumar et al. (2015), and the Neural Reasoner (Peng et al., 2015). MemN2N has since been shown to perform well on some real-world tasks (Hill et al., 2015).

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# A EXTENSIONS TO MEMORY NETWORKS

Memory Networks Weston et al. (2014) are a promising class of models, shown to perform well at QA, that we can apply to our tasks. They consist of a memory $\mathbf{m}$ (an array of objects indexed by $\mathbf{m}_i$) and four potentially learnable components $I$, $G$, $O$ and $R$ that are executed given an input:

I: (input feature map) – convert input sentence $x$ to an internal feature representation $I(x)$.

G: (generalization) – update the current memory state $\mathbf{m}$ given the new input: $\mathbf{m}_i = G(\mathbf{m}_i, I(x), \mathbf{m}),\ \forall i$.

O: (output feature map) – compute output $o$ given the new input and the memory: $o = O(I(x), \mathbf{m})$.

R: (response) – finally, decode output features $o$ to give the final textual response to the user: $r = R(o)$.

Potentially, component $I$ can make use of standard pre-processing, e.g., parsing and entity resolution, but the simplest form is to do no processing at all. The simplest form of $G$ is store the new incoming example in an empty memory slot, and leave the rest of the memory untouched. Thus, in Weston et al. (2014) the actual implementation used is exactly this simple form, where the bulk of the work is in the $O$ and $R$ components. The former is responsible for reading from memory and performing inference, e.g., calculating what are the relevant memories to answer a question, and the latter for producing the actual wording of the answer given $O$.

The $O$ module produces output features by finding $k$ supporting memories given $x$. They use $k = 2$. For $k = 1$ the highest scoring supporting memory is retrieved with:

$$o_1 = O_1(x, \mathbf{m}) = \underset{i=1,\dots,N}{\arg\max}\ s_O(x, \mathbf{m}_i) \tag{1}$$

where $s_O$ is a function that scores the match between the pair of sentences $x$ and $\mathbf{m}_i$. For the case $k = 2$ they then find a second supporting memory given the first found in the previous iteration:

$$o_2 = O_2(q, \mathbf{m}) = \underset{i=1,\dots,N}{\arg\max}\ s_O([x, \mathbf{m}_{o_1}], \mathbf{m}_i) \tag{2}$$

where the candidate supporting memory $\mathbf{m}_i$ is now scored with respect to both the original input and the first supporting memory, where square brackets denote a list. The final output $o$ is $[x, \mathbf{m}_{o_1}, \mathbf{m}_{o_2}]$, which is input to the module $R$.

Finally, $R$ needs to produce a textual response $r$. While the authors also consider Recurrent Neural Networks (RNNs), their standard setup limits responses to be a single word (out of all the words seen by the model) by ranking them:

$$r = R(q, w) = \text{argmax}_{w \in W}\ s_R([x, \mathbf{m}_{o_1}, \mathbf{m}_{o_2}], w) \tag{3}$$

where $W$ is the set of all words in the dictionary, and $s_R$ is a function that scores the match.

The scoring functions $s_O$ and $s_R$ have the same form, that of an embedding model:

$$s(x, y) = \Phi_x(x)^\top U^\top U \Phi_y(y). \tag{4}$$

where $U$ is a $n \times D$ matrix where $D$ is the number of features and $n$ is the embedding dimension. The role of $\Phi_x$ and $\Phi_y$ is to map the original text to the $D$-dimensional feature space. They choose a bag of words representation, and $D = 3|W|$ for $s_O$, i.e., every word in the dictionary has three different representations: one for $\Phi_y(.)$ and two for $\Phi_x(.)$ depending on whether the words of the input arguments are from the actual input $x$ or from the supporting memories so that they can be modeled differently.

They consider various extensions of their model, in particular modeling write time and modeling unseen words. Here we only discuss the former which we also use. In order for the model to work on QA tasks over stories it needs to know which order the sentences were uttered which is not available in the model directly. They thus add extra write time extra features to $S_O$ which take on the value 0 or 1 indicating which sentence is older than another being compared, and compare triples of pairs of sentences and the question itself. Training is carried out by stochastic gradient descent using supervision from both the question answer pairs and the supporting memories (to select $o_1$ and $o_2$). See Weston et al. (2014) for more details.